

International Islamic University Chittagong Department of
Electrical and Electronic Engineering

(Project/Thesis Proposal)

Application for the Approval of Project/Thesis Proposal

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4. Date of Assigning Supervisor:

09-09-2025

5. Date of Proposal Submission:

6. Tentative Title:

Performance Comparison of 1-Bit Magnitude Comparator Using Static, TG, PTL, and GDI
Logic Styles

7. Introduction

The continuous scaling of CMOS technology has enabled the development of high-performance and energy-efficient digital integrated circuits. Digital magnitude comparators are fundamental building blocks in modern digital systems because they determine the relative relationship between two binary inputs. A 1-bit magnitude comparator produces three possible outputs indicating whether input A is greater than input B , equal to input B , or less than input B . Such circuits are widely used in arithmetic logic units (ALUs), processors, control systems, digital signal processing units, memory systems, and other VLSI applications.

The implementation of a comparator at the transistor level is strongly influenced by the selected logic style. Static CMOS logic provides reliable operation, good noise margins, and full voltage swing, but it may require a relatively large number of transistors. Transmission Gate (TG) logic can reduce transistor count and provide efficient signal transmission. Pass Transistor Logic (PTL) offers compact circuit implementation and potentially lower power consumption,

although threshold-voltage loss and reduced voltage swing can affect its performance. Gate Diffusion Input (GDI) logic provides another low-complexity implementation technique that can reduce transistor count, power consumption, and circuit area.

Therefore, selecting an appropriate logic style requires a systematic comparison of important performance parameters rather than considering a single parameter. This research proposes the design and performance comparison of a 1-bit magnitude comparator implemented using Static CMOS, Transmission Gate (TG), Pass Transistor Logic (PTL), and Gate Diffusion Input (GDI) logic styles. The designs will be implemented at the transistor level and evaluated using simulation. The comparison will focus on propagation delay, average power consumption, transistor count, and Power Delay Product (PDP). The study is expected to identify the most suitable logic style for efficient 1-bit magnitude comparator design for low-power and high-performance VLSI applications.

8. Literature Review

CMOS logic design has been extensively investigated to achieve lower power consumption, higher operating speed, reduced transistor count, and improved area efficiency. Conventional Static CMOS remains one of the most widely used logic implementation techniques because of its robust operation, full voltage swing, and good noise immunity. However, complementary pull-up and pull-down networks can result in a comparatively higher transistor count for certain logic functions.

Alternative logic styles have been introduced to overcome some of the limitations of conventional CMOS implementation. Transmission Gate (TG) logic uses parallel NMOS and PMOS devices to transmit both logic levels effectively. This approach can provide efficient switching and reduce circuit complexity for selected digital functions. However, the use of additional control signals and transistor sizing can influence overall circuit performance.

Pass Transistor Logic (PTL) reduces the number of transistors by using transistors as controlled switches. PTL-based circuits can achieve compact implementation and lower switching activity. However, threshold-voltage degradation, reduced voltage swing, and signal restoration requirements may affect the reliability and speed of the circuit.

Gate Diffusion Input (GDI) is another low-power logic design technique that modifies the conventional CMOS structure by applying different input signals to the gate and diffusion terminals of MOS transistors. GDI can significantly reduce transistor count and switching activity for several logic functions. Nevertheless, its performance can depend on the technology, transistor sizing, input configuration, and output restoration requirements.

Magnitude comparators have been implemented using different CMOS logic techniques to achieve a balance among speed, power, area, and circuit complexity. Previous research generally emphasizes one or two performance parameters, while a comprehensive comparison of Static CMOS, TG, PTL, and GDI implementations of the same 1-bit magnitude comparator under identical simulation conditions is less frequently emphasized.

This research therefore focuses on implementing the same 1-bit magnitude comparator us-

ing four different logic styles and evaluating them under consistent design and simulation conditions. The comparison will provide a clearer understanding of the trade-offs among power consumption, propagation delay, transistor count, and Power Delay Product.

9. Problem Statement / Research Gap

Although several logic styles are available for implementing digital comparator circuits, each technique has different advantages and limitations. Static CMOS generally provides reliable full-swing operation and good noise margins, but its complementary transistor structure can increase transistor count and switching capacitance. TG logic can provide efficient signal transmission but may introduce additional control complexity. PTL can reduce transistor count and circuit complexity, but threshold-voltage degradation may affect the output signal level. GDI can achieve highly compact implementations and low switching activity, but its performance and voltage-level behavior may vary depending on circuit topology and operating conditions.

Furthermore, comparing different logic styles using different circuit structures, simulation conditions, or performance criteria can make it difficult to determine which technique provides the best overall implementation for a 1-bit magnitude comparator. A direct and systematic comparison of Static CMOS, TG, PTL, and GDI logic styles using the same comparator functionality and consistent evaluation criteria is therefore required.

The research gap addressed in this work is the lack of a unified performance comparison of these four logic styles for a 1-bit magnitude comparator under equivalent design and simulation conditions. The study will investigate the trade-offs among propagation delay, average power consumption, transistor count, and Power Delay Product. The findings will help identify the logic style that provides the most suitable balance between speed, power efficiency, and circuit complexity.

10. Objectives with Specific Aims

General Objective

To design and perform a comparative analysis of a 1-bit magnitude comparator using Static CMOS, Transmission Gate (TG), Pass Transistor Logic (PTL), and Gate Diffusion Input (GDI) logic styles.

Specific Objectives

1. To design transistor-level 1-bit magnitude comparator circuits using Static CMOS, TG, PTL, and GDI logic styles.
2. To simulate and verify the functional operation of all four comparator implementations under identical simulation conditions.

3. To measure and analyze propagation delay, average power consumption, transistor count, and Power Delay Product (PDP) for each logic style.
4. To compare the performance of Static CMOS, TG, PTL, and GDI implementations and identify the most suitable logic style based on the obtained performance trade-offs.

11. Proposed Methodology

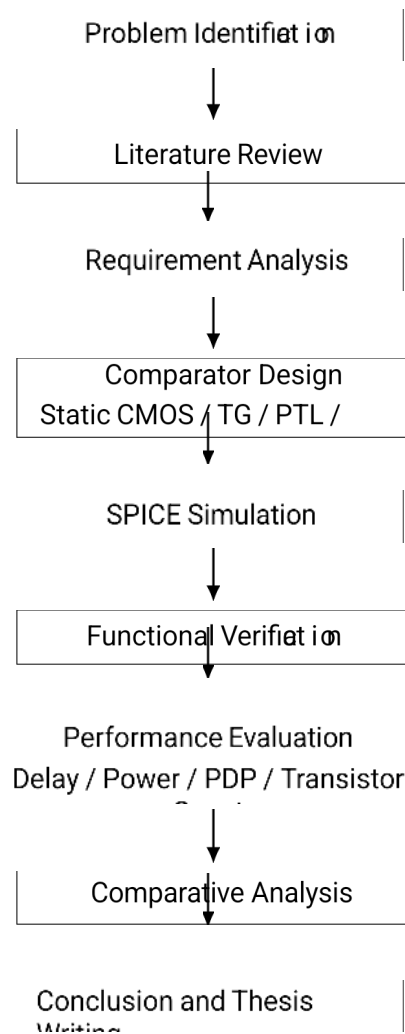
The proposed research will follow a systematic design, simulation, verification, and performance evaluation procedure. First, relevant literature on 1-bit magnitude comparators and CMOS logic styles will be reviewed to identify existing design approaches and research gaps. Based on the literature review, the required design specifications and performance parameters will be established.

A 1-bit magnitude comparator will then be designed using four different logic styles: Static CMOS, Transmission Gate (TG), Pass Transistor Logic (PTL), and Gate Diffusion Input (GDI). The same Boolean functionality and equivalent input conditions will be maintained for all four implementations to ensure a fair comparison.

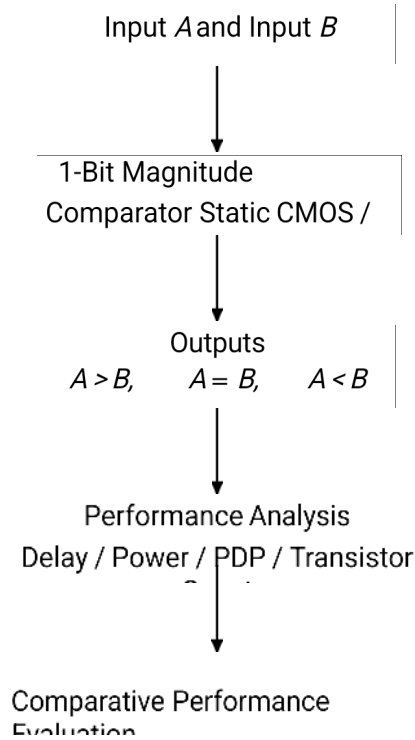
The transistor-level circuits will be implemented and simulated using an appropriate SPICE-based simulation environment. Functional verification will be performed for all possible combinations of the two binary inputs. After successful functional verification, the performance of each implementation will be evaluated using propagation delay, average power consumption, transistor count, and Power Delay Product (PDP).

Finally, the obtained results will be organized and compared using suitable tables and graphs. The comparative analysis will identify the strengths and limitations of each logic style and determine which implementation provides the most appropriate balance between speed, power consumption, and circuit complexity.

Research Framework



Block Diagram of the Proposed System



12. Gantt Chart

The proposed Final Year Project/Thesis will be completed over a period of 12 months. The planned activities include topic selection, literature review, proposal preparation, methodology development, circuit design, simulation, testing, performance evaluation, thesis writing, revision, and final submission.

Activities	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Topic Selection and Problem Identification	•	•										
Literature Review	•	•	•	•								
Proposal Preparation and Presentation		•	•									
Requirement Analysis and Methodology Development			•	•	•							
Comparator Circuit Design				•	•	•						
SPICE Simulation					•	•	•					
Testing and Functional Verification						•	•	•				
Performance Analysis							•	•	•			
Comparative Evaluation								•	•	•		
Thesis Writing and Documentation									•	•	•	
Thesis Revision											•	•
Final Submission, Presentation and Defense												•

13. References

References

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14. Justification of the Problem as Complex Engineering Problem

The proposed research addresses a complex engineering problem because it involves transistor-level digital VLSI design, analysis of multiple CMOS logic styles, SPICE-based simulation, functional verification, and multi-parameter performance evaluation.

The research requires balancing several competing engineering requirements, including propagation delay, average power consumption, transistor count, Power Delay Product, volt-age swing, and reliable logic operation. The four investigated logic styles have different circuit structures and electrical characteristics, making the selection of an optimal architecture a multi-dimensional engineering decision.

The proposed work addresses the following Washington Accord Graduate Attributes (WA):

- **WA1 – Engineering Knowledge:** Application of knowledge of CMOS devices, digital logic, VLSI design, and circuit analysis.
- **WA2 – Problem Analysis:** Identification and analysis of power, delay, transistor count, and circuit complexity trade-offs.
- **WA3 – Design and Development of Solutions:** Design and development of four transistor-level 1-bit magnitude comparator architectures.
- **WA4 – Investigation:** Simulation, testing, measurement, and comparative analysis of circuit performance.
- **WA5 – Modern Tool Usage:** Use of SPICE-based simulation and electronic design automation tools for circuit development and evaluation.
- **WA12 – Life-long Learning:** Review and application of current research in low-power CMOS and VLSI circuit design.

15. Budget (if any)

Item	Cost (BDT)
Software	0
Internet	1500
Printing and Binding	2000
Miscellaneous	1500
Total	5000

Signature of the
Student

Signature of the
Supervisor

Signature of
the
Convener
Thesis/

Project
Commi
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